# Ali Reza Ibrahimzada

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## About Me

I received my CS PhD from Siebel School of Computing and Data Science at the University of Illinois Urbana-Champaign. During my four years at UIUC, I worked on code translation using software engineering and machine learning techniques. My PhD research studied the effectiveness of LLMs in code translation [C6] and built early tools for neuro-symbolic [C1, C5] and agentic [C2, C3] repository-level code translation and validation.

## Education

08/22 – 08/26 **Ph.D. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Neuro-Symbolic Code Translation and Validation*

08/22 – 12/24 **M.Sc. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Bridging the Gap between Testing and Debugging through Explainable Deep Oracles*

09/18 – 07/22 **B.Sc. Computer Engineering, Marmara University, Istanbul, Turkey**
GPA: 3.98 / 4.00
Advisor: Mehmet Kadir Baran
Thesis: *Depth Estimation of Stereo Images using Deep Learning*

## Professional Experience

08/22 – 08/26 **Research Assistant,** University of Illinois Urbana-Champaign, IL, USA
Advised by: Reyhaneh Jabbarvand
Topic: Code LLMs for Source Code Generation, Understanding and Translation
Responsibilities:

- Performing research in the intersection of large language models and software engineering.
- Evaluating large language models in code translation.
- Building neuro-symbolic tools for unified translation and validation of repository-level code.
- Creating agentic architectures for translation validation and repair.
- Developing agentic workflows for end-to-end whole-repository translation and validation.

05/25 – 08/25 **Applied Scientist Intern,** Amazon Web Services, VA, USA
Hosted by: Brandon Paulsen and Joey Dodds
Topic: LLM Agents for Autonomous Translation Validation and Repair
Responsibilities:

- Building MatchFixAgent, a language-agnostic and agentic technique for translation validation and repair.
- Creating benchmarks from existing techniques and evaluating MatchFixAgent.
- Performing cost analysis of MatchFixAgent against existing language-specific techniques.

## Professional Experience (continued)

05/24 – 08/24 **Research Intern,** IBM Research, NY, USA
Hosted by: Saurabh Sinha and Raju Pavuluri
Topic: End-to-End Repository-level Code Translation and Validation
Responsibilities:

- Developing AlphaTrans, the first end-to-end neuro-symbolic tool for repository-level code translation and validation.
- Creating a benchmark of real-world Java repositories and evaluating AlphaTrans on them.
- Developing a language interoperability approach for validating translations in isolation.

05/23 – 08/23 **Research Intern,** IBM Research, NY, USA
Hosted by: Saurabh Sinha and Raju Pavuluri
Topic: Evaluating LLMs in Code Translation and Bug Analysis
Responsibilities:

- Performing a large-scale empirical study on LLM-based code translation.
- Performing a large-scale human study on translation bugs generated by LLMs.
- Comparing LLMs with transpilers in code translation.

05/21 – 08/22 **Research Intern,** University of Illinois Urbana-Champaign, IL, USA
Hosted by: Reyhaneh Jabbarvand
Topic: Neural and Interpretable Test Oracles using Deep Learning
Responsibilities:

- Building SEER, a deep neural model for test oracle problem.
- Evaluating SEER on the Defects4J benchmark.
- Comparing SEER with traditional mutation-based techniques.

06/19 – 02/22 **Research Intern,** Istanbul Technical University, Istanbul, Turkey
Hosted by: Ali Cakmak
Topic: Machine Learning
Responsibilities:

- Research on using machine learning techniques for scalable classification of living organisms.
- Developing machine learning techniques for predicting future COVID mutants.
- Developing tools for predicting student success using machine learning.

## Research Publications

### Conference Proceedings

[C1] K. Ke, **A. R. Ibrahimzada**, R. Pan, S. Sinha, and R. Jabbarvand, "Mockingbird: Context-aware repository-level code translation with isolated validation," in *2027 IEEE/ACM 49th International Conference on Software Engineering (ICSE)*, 2027. URL: https://arxiv.org/abs/2511.21878

[C2] **A. R. Ibrahimzada**, B. Paulsen, D. Kroening, and R. Jabbarvand, "Recodeagent: A multi-agent workflow for language-agnostic translation and validation of large-scale repositories," in *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering*, ser. ASE '26, Munich, Germany: Association for Computing Machinery, 2026, pp. 2409–2421, ISBN: 9798400728822. DOI: 10.1145/3832783.3837486

[C3] **A. R. Ibrahimzada**, B. Paulsen, R. Jabbarvand, J. Dodds, and D. Kroening, "MatchFixAgent: Language-agnostic autonomous repository-level code translation validation and repair," in *Proceedings of the 43rd International Conference on Machine Learning*, T. Zhang, M. Dudik, M. Jaggi, A. Agarwal, S. Li, D. Schuurmans, J. Zhu, F. Berkenkamp, H. Dong, and A. Bietti, Eds., ser. Proceedings of Machine

Learning Research, vol. 306, PMLR, Jun. 2026, pp. 49 428–49 451. URL: https://proceedings.mlr.press/v306/ibrahimzada26a.html

[C4] **A. R. Ibrahimzada**, Y. Chen, R. Rong, and R. Jabbarvand, "Challenging bug prediction and repair models with synthetic bugs," in *2025 IEEE International Conference on Source Code Analysis & Manipulation (SCAM)*, 2025, pp. 133–144. DOI: 10.1109/SCAM67354.2025.00021

[C5] **A. R. Ibrahimzada**, K. Ke, M. Pawagi, M. S. Abid, R. Pan, S. Sinha, and R. Jabbarvand, "Alphatrans: A neuro-symbolic compositional approach for repository-level code translation and validation," FSE, vol. 2, New York, NY, USA: Association for Computing Machinery, Jun. 2025. DOI: 10.1145/3729379

[C6] R. Pan*, **A. R. Ibrahimzada***, R. Krishna, D. Sankar, L. P. Wassi, M. Merler, B. Sobolev, R. Pavuluri, S. Sinha, and R. Jabbarvand, "Lost in translation: A study of bugs introduced by large language models while translating code," in *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*, ser. ICSE '24, Lisbon, Portugal: Association for Computing Machinery, 2024, ISBN: 9798400702174. DOI: 10.1145/3597503.3639226

[C7] **A. R. Ibrahimzada**, "Program decomposition and translation with static analysis," in *Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings*, ser. ICSE-Companion '24, Lisbon, Portugal: Association for Computing Machinery, 2024, pp. 453–455, ISBN: 9798400705021. DOI: 10.1145/3639478.3641226

[C8] **A. R. Ibrahimzada**, Y. Varli, D. Tekinoglu, and R. Jabbarvand, "Perfect is the enemy of test oracle," in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ser. ESEC/FSE 2022, Singapore, Singapore: Association for Computing Machinery, 2022, pp. 70–81, ISBN: 9781450394130. DOI: 10.1145/3540250.3549086

### Journal Articles

[J1] **A. R. Ibrahimzada**, K. Kosif, A. S. Gulsen, Y. S. Yaglica, and A. Cakmak, "Predicting student grades via adaptive multi-level learning models," *Scientific Reports*, 2026. URL: https://doi.org/10.1038/s41598-026-55750-z

[J2] A. Cakmak, H. Ayaz, S. Arıkan, **A. R. Ibrahimzada**, Ş. Demirkol, D. Sönmez, M. T. Hakan, S. T. Sürmen, C. Horozoğlu, M. B. Doğan, Ö. Küçükhüseyin, C. Cacına, B. Kıran, Ü. Zeybek, M. Baysan, and İ. Yaylım, "Predicting the predisposition to colorectal cancer based on snp profiles of immune phenotypes using supervised learning models," *Medical & Biological Engineering & Computing*, vol. 61, no. 1, pp. 243–258, Jan. 2023, ISSN: 1741-0444. DOI: 10.1007/s11517-022-02707-9

[J3] G. N. Sohsah, **A. R. Ibrahimzada**, H. Ayaz, and A. Cakmak, "Scalable classification of organisms into a taxonomy using hierarchical supervised learners," *Journal of Bioinformatics and Computational Biology*, vol. 18, no. 05, p. 2 050 026, 2020, PMID: 33125294. DOI: 10.1142/S0219720020500262 eprint: https://doi.org/10.1142/S0219720020500262.

### Thesis

[T1] **A. R. Ibrahimzada**, *Neuro-symbolic code translation and validation*, 2026.

[T2] **A. R. Ibrahimzada**, *Bridging the gap between testing and debugging through explainable deep oracles*, 2024.

## Awards

2024 **Nomination for the Two Sigma PhD Fellowship**, University of Illinois Urbana-Champaign

**3rd in Student Research Competition @ ICSE'24**, ACM

**Travel Grant to attend ICSE'24 in Lisbon, Portugal**, NSF

2022 **Travel Grant to attend ESEC/FSE'22 in Singapore, Singapore**, ACM

## Awards (continued)

- **Valedictorian of Class of 2022**, Marmara University
- **Ray Ozzie Computer Science Fellowship**, University of Illinois Urbana-Champaign
- **Best Senior Graduation Project of the Year**, Marmara University

2019 **Academic Achievement Scholarship**, Marmara University

2017 **Valedictorian of High School**, KEN

2015 **Travel Grant to attend QUEST'15 in Lucknow, India**, KEN

## Talks

ASE'26 ReCodeAgent: A Multi-agent Workflow for Language-Agnostic Translation and Validation of Large-Scale Repositories, Munich, Germany

ICML'26 MatchFixAgent: Language-Agnostic Autonomous Repository-Level Code Translation Validation and Repair, Seoul, South Korea

SCAM'25 Challenging Bug Prediction and Repair Models with Synthetic Bugs, Auckland, New Zealand

FSE'25 AlphaTrans: A Neuro-Symbolic Compositional Approach for Repository-Level Code Translation and Validation, Trondheim, Norway

ICSE'24 Lost in Translation: A Study of Bugs Introduced by Large Language Models while Translating Code, Lisbon, Portugal

FSE'22 Perfect Is the Enemy of Test Oracle, Singapore, Singapore

## Skill Stack

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|---|---|
| Programming Languages | Python, Java, JavaScript, Rust, C, OCaml |
| Common | Docker, Vim, Git, SQL |
| Machine Learning | PyTorch, TensorFlow, vLLM, Codex, Claude Code |

## Service

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|---|---|
| Program Committee | NeurIPS'26 (Sydney, Australia), ICML'26 (Seoul, South Korea), ICLR'26 (Rio de Janeiro, Brazil), ICLR'25 (Singapore, Singapore), MSR'25 (Ottawa, Canada), MSR'24 (Lisbon, Portugal) |
| Artifact Evaluation | ICSE'25 (Ottawa, Canada) |
| Organizing | ReCode@ICSE'27 (Dublin, Ireland), ReCode@ICSE'26 (Rio de Janeiro, Brazil) |

## Teaching Experience

Fall 2025 CS 421 - Programming Languages and Compilers w/ Elsa Gunter

Spring 2025 CS 447 - Natural Language Processing w/ Julia Hockenmaier

## References

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